

Assignment No:5

Title of the Assignment: Implementation of High Performance Computing (HPC) Application for AI/ML using TensorFlow Distributed Training

Objective of the Assignment: To implement a high-performance computing (HPC) based AI/ML model using parallel processing techniques to improve training speed and efficiency.

Theory:

High Performance Computing (HPC) refers to the use of powerful computing resources such as multi-core processors, GPUs, and distributed systems to solve complex computational problems efficiently. In the domain of Artificial Intelligence and Machine Learning, HPC plays a crucial role in handling large datasets and training complex models within a reasonable time.

HPC systems use **parallel computing**, where multiple computations are performed simultaneously. This is achieved using techniques such as:

- **Data Parallelism:** Splitting data across multiple processors and performing the same operation on each subset.
- **Model Parallelism:** Dividing the model across different processors for simultaneous computation.

In AI/ML, training deep learning models involves large matrix operations and iterative computations, which are computationally expensive. HPC accelerates this process using:

- **GPU Acceleration:** GPUs perform thousands of parallel operations, making them ideal for deep learning tasks.
- **Distributed Computing:** Multiple machines (nodes) work together to train a model faster.
- **Batch Processing:** Data is processed in batches to optimize memory usage and speed.

Frameworks like **TensorFlow** and **Keras** provide built-in support for HPC through distributed training strategies such as:

- `MirroredStrategy()` for single-machine multi-GPU training
- `MultiWorkerMirroredStrategy()` for multi-node distributed training

These strategies replicate the model across devices and synchronize updates, enabling efficient parallel training.

HPC also improves:

- **Scalability:** Ability to handle increasing data size and model complexity
- **Performance:** Faster training and reduced execution time
- **Resource Utilization:** Efficient use of CPU/GPU resources

However, HPC systems may require specialized hardware, high setup cost, and technical expertise to manage distributed environments.

Algorithm

1. Load dataset (MNIST)
2. Normalize input data
3. Initialize distributed strategy
4. Build model inside strategy scope
5. Compile model
6. Train model using parallel execution
7. Evaluate performance

Conclusion

Parallel computing in HPC significantly enhances the performance of AI/ML models by utilizing multiple processing units. This approach is essential for handling large-scale machine learning problems.